



**KAZAKH-BRITISH
TECHNICAL
UNIVERSITY**

**JSC «Kazakh British Technical University»
School of IT and Engineering**

**APPROVED BY
Dean of SITE**

Azamat Imanbayev

2025



SYLLABUS

Discipline: Basics of Information Systems

Term: Fall 2025

Instructors' full name: Assel Smaiyl Maralbaikyzy, Aslan Medeuov

Personal Information about the Instructor	Time and place of classes		Contact information	
	Lessons	Office Hours	Tel.	e-mail
Assel Smaiyl Maralbaikyzy Ph.D	According to the schedule	According to the schedule	Telegram @smaiyl_assel	a.smaiyl@kbtu.kz
Aslan Medeuov Senior lecturer	According to the schedule	According to the schedule		a.medeuov@kbtu.kz

COURSE DURATION: 3 credits, 15 weeks, 45 class hours

MS Teams code

GENERAL COURSE AIMS:

The goal of teaching discipline "Basics of Information Systems" is to:

1. Understand the information systems for business and their components.
2. Know the use of information systems for achieving strategic advantage.
3. Know the latest hardware and software resources, telecommunication networks, and data resource management concepts.
4. Understand the planning fundamentals and implementation challenges associated with business/IT strategies.

COURSE DESCRIPTION

This course provides an overview of using information systems to support management activities in different business organizations. Increasingly, IT is being utilized to achieve operational excellence, implement business strategies, and gain competitive advantage, not merely to support

business transactions. This course will cover recent topics, including the role of information systems, hardware, software, telecommunications, managing data resources, security elements, and systems development in business operations.

COURSE OBJECTIVES

1. Understand the foundation concepts of information systems for business.
2. Realize the use of information systems in today's competitive environment to achieve strategic advantage.
3. Develop an awareness of the latest hardware and software resources and telecommunication networks.
4. Realize the importance of data resource management
5. Know the planning fundamentals and implementation challenges associated with business/IT strategies and understand the security, societal, and ethical challenges of IT

COURSE OUTCOMES

1. Ability to illustrate the main concepts of Information Systems and describe their main roles in organizations.
2. Explain the different types of management information systems and their uses.
3. Recognize the new technologies transforming businesses and their influence in the near future.
4. Solve business problems using information technology components and applications.
5. Evaluate the security, societal, and ethical challenges of IT.

COURSE POST REQUISITES: -

Knowledge and skills obtained during the study of the course Basics to Information Systems are used in the following courses: Foundations of Business for Information Systems, Management Information Systems, Enterprise Resource Planning, and Emerging Technologies for Business.

LITERATURE

1. Laudon, K. C. Management Information Systems: Managing the Digital Firm / K. C. Laudon, J. P. Laudon. — 17th ed. — Harlow : Pearson, 2023. — 624 p.
2. O'Brien, J. A. Management Information Systems / J. A. O'Brien, G. M. Marakas. — 11th ed. — New York : McGraw-Hill Education, 2019. — 720 p.
3. Stair, R. Principles of Information Systems / R. Stair, G. Reynolds. — 13th ed. — Boston : Cengage Learning, 2020. — 640 p.
4. Pearlson, K. E. Managing and Using Information Systems: A Strategic Approach / K. E. Pearlson, C. S. Saunders, D. F. Galletta. — 8th ed. — Hoboken : Wiley, 2023. — 368 p.
5. Turban, E. Information Technology for Management: Advancing Sustainable, Profitable Business Growth / E. Turban, C. Pollard, G. Wood. — 12th ed. — Hoboken : Wiley, 2018. — 672 p.
6. Bocij, P. Business Information Systems: Technology, Development and Management / P. Bocij, A. Greasley, S. Hickie. — 6th ed. — Harlow : Pearson, 2019. — 720 p.
7. Chaffey, D. Business Information Management / D. Chaffey, G. White. — 3rd ed. — Harlow : Pearson, 2011. — 728 p.
8. Zwass, V. Foundations of Information Systems / V. Zwass. — New York : McGraw-Hill, 1997. — 720 p.
9. Alter, S. Information Systems: The Foundation of E-Business / S. Alter. — Harlow : Pearson, 2002. — 650 p.

10. Laudon, K., Laudon, J. Information Systems. Management in the Digital Age : textbook / K. Laudon, J. Laudon. — Moscow : Williams, 2022. — 800 p.
11. Oikhman, E. G., Popov, E. V. Information Systems and Technologies in Economics : textbook / E. G. Oikhman, E. V. Popov. — Moscow : Finance and Statistics, 2019. — 432 p.
12. Telnov, Yu. F. Information Systems and Technologies in Management : textbook / Yu. F. Telnov. — Moscow : Yurayt, 2021. — 410 p.
13. Glushkov, V. M. Introduction to Information Systems : textbook / V. M. Glushkov. — Moscow : Nauka, 2020. — 356 p.

Reading Textbook 1

1. James A. O'Brien; George M. Marakas, "Management Information Systems", McGraw-Hill Irwin, ISBN 978-0-07-337681-3

Reference Textbook 2

2. Kenneth C. Laudon; Jane Price Laudon, (2020), "Management information systems: managing the digital firm" Pearson Education, ISBN: 9780135191798, 0135191793.

COURSE ASSESSMENT CRITERIA

Assessment occurs continuously throughout the course. The evaluation will be based on the levels of (maximums in %):

Type of activity	Final scores
SIS (Projects, 4 total)	28
Midterm/Endterm	32
Final exam	40
Total	100

TASKS

for students' independent study (SIS)

Week	SIS	Cost (in points)
4	SIS1	7
7	SIS2	7
11	SIS3	7
14	SIS4	7
	Total	28

COURSE CALENDAR

Week	Class work			SIS
	Topic	Lectures (hours)	Practice (hours)	SIS defense
1	Introduction to Information Systems	3	0	

2	Components of IS	3	0	
3	IS for Strategic Advantage	3	0	
4	Computer Hardware: Architecture & Sizing	3	0	SIS1
5	Computer Hardware & Peripherals	3	0	
6	Computer Software	3	0	
7	Data Resource Management: Governance & Quality	3	0	SIS2
8	Midterm Exam	3	0	
9	Data Resource Management (Databases)	3	0	
10	Telecommunications & Networks	3	0	
11	Security Foundations: Risk & Controls	3	0	SIS3
12	Security & Ethical Challenges (Threats, Privacy, Ethics)	3	0	
13	Emerging Technologies in IS	3	0	
14	Developing Business/IT Strategies: Planning & Implementation	3	0	SIS4
15	End term Exam	3	0	
16-17	Final Exam			

Course assessment schedule

No	Assessment criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
1	Attendance / participation	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	0 %
2	SIS				*			*				*			*		28%
3	Midterm/ End-term								*							*	32%
4	Final exam																40%
	Total																100%

Class sessions – will be a mixture of information, discussion and practical application of skills.

Practice – will reinforce the students knowledge by practical application of lectured materials.

In-class assessment – will prepare students for their mid-term and final assessment and identify the competence level they have achieved on a related subject matter, the aim being to diagnose potential

discrepancies in students' understanding and performance in order to make specific adjustments to the course content and procedures and/or to assign additional assignments to certain individuals or the whole group.

Home assignments – will consolidate the concepts and materials taken during in-class activities, help students to expand the content through diverse background resources and/or practise certain skill areas; they will also develop the students' ability to work individually in exploring and examining related issues.

SIS (Student Independent Study) – comprises group Project to be done by students on the independent basis. Students are supposed to use knowledge and skills acquired in class to do the project. Assistance and advice will be provided by teachers during office hours.

TSIS (Teacher Supervised Student Independent Study) – student self-made project.

End-term test – a diagnostic test used to identify the students' progress, their strengths and weaknesses, intended to force student to prepare for Final Exam. It includes computer based test.

Final examination – 1) an attainment test designed to identify how successful the students have been achieving objectives.

Grading policy:

Intermediate attestations (on 8th and 15th week) join topics of all lectures, practice, laboratories, SIS, TSIS and materials for reading discussed to the time of attestation. Maximum number of points within attendance, activity, SIS, TSIS and laboratories for each attestation is 30 points.

Final exam joins and generalizes all course materials, is conducted in the complex form with quiz and problem. Final exam duration is 180 min. Maximum number of points is 40. At the end of the semester you receive overall total grade (summarized index of your work during semester) according to conventional KBTU grade scale.

ATTENTION!

- 1) If student missed more than **30% of lessons student receives «F (Fail)» grade;**
 - 2) If for two attestations student receives 29.4 or less points, this student is not accepted to final exam and for all course he (she) receives **«F (Fail)» grade;**
 - 3) If student receives 9.4 or less points on final exam, then independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade;**
- In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade.**

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation". Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.
6. Make up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

Grade	Achievement percentage	Assessment criterion
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«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks; Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
	B	80-84%	
	B-	75-79%	
	C+	70-74%	
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.
	C-	60-64%	
	D+	55-59%	
	D	50-54%	

Assistant-Professor of SITE
Senior-Lecturer of SITE

Dr. Assel Smaiyl
Aslan Medeuov

Minutes # 1 of School of Information Technology and Engineering meeting on August 18, 2025